Code No: 133BM **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, April/May - 2018
OBJECT ORIENTED PROGRAMMING THROUGH JAVA
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a) What is abstract class? Give example. [2]
b) Explain the use of 'for' statement in Java with an example. [3]
c) Define a Package? What is its use in java? Explain. [2]
d) List out the benefits of Stream oriented I/O. [3]
e) How do we start and stop a thread? [2]
f) Write the complete life cycle of a thread. [3]
g) What is the benefit of Generics in Collections Framework? [2]
h) Differentiate between Enumeration and Iterator interface. [3]
i) What are the limitations of AWT? [2]
j) Why do applet classes need to be declared as public? [3]

PART-B

(50 Marks)

2.a) What is inheritance and how does it help to create new classes quickly.
b) Describe different levels of access protection available in Java. [5+5]

OR

3.a) List the primitive data types available in Java and explain.
b) What is polymorphism? Explain different types of polymorphisms with examples. [5+5]

4.a) What is an interface? What are the similarities between interfaces and classes?
b) How can you extend one interface by the other interface? Discuss. [5+5]

OR

5.a) Discuss about CLASSPATH environment variables.
b) Discuss the different levels of access protection available in Java. [5+5]

6.a) What are advantages of using Exception handling mechanism in a program?
b) Write a java program that demonstrates how certain exception types are not allowed to be thrown. [5+5]

OR

7.a) What are the different ways that are possible to create multiple threaded programs in java? Discuss the differences between them.
b) Write a program to create four threads using Runnable interface. [5+5]

8.a) What is Java Collections Framework? List out some benefits of Collections framework and explain.

b) What is the importance of hashCode() and equals() methods? [5+5]

OR

9.a) What are the common algorithms implemented in Collections Framework? Discuss.

b) What is difference between ArrayList and LinkedList in collection framework? Explain. [5+5]

10.a) What is an applet? Explain the life cycle of Applet with a neat sketch.

b) Write the applets to draw the Cube and Cylinder shapes. [5+5]

OR

11.a) What is an Layout manager? Explain different types of Layout managers.

b) Write a program to create a frame window that responds to key strokes. [5+5]

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